HWRS 561b: Physical Hydrogeology II

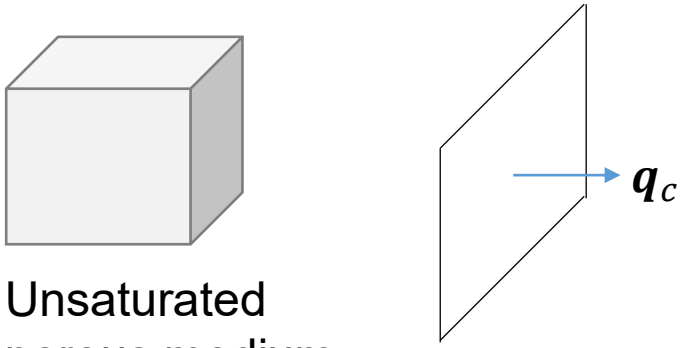
Unsat solute transport

Agenda:

1. ADE for aqueous unsat solute transport
2. Vapor phase transport

Reading: FBK 4.7-4.13

Solute transport under saturated flow



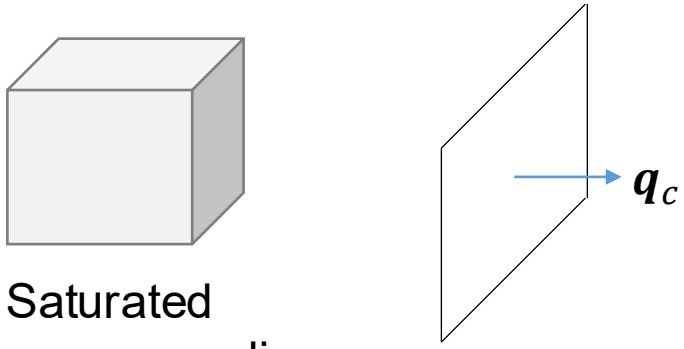
Unsaturated porous medium

$$\boldsymbol{q}_c = \theta(\boldsymbol{v}C - \boldsymbol{D}\nabla C),$$

- $\boldsymbol{q}_c$ is the solute flux vector.
- $\boldsymbol{v}$ is the velocity field.
- $\boldsymbol{D}$ is a tensor that depends on the velocity field, soil moisture, and molecular diffusion in the porous medium.

$$\boldsymbol{D} = \alpha_T(\theta)|\boldsymbol{v}|\mathbf{I} + (\alpha_L(\theta) - \alpha_T(\theta))\frac{\boldsymbol{v}\boldsymbol{v}}{|\boldsymbol{v}|} + w(\theta)D_0\mathbf{I}$$

- Dispersivities and tortuosity factor are functions of water content.
- Often, tortuosity factor is approximated as $w(\theta) = \theta^{7/3}/\theta_s^2$. Dependence of dispersivity on water content is difficult to determine.



Saturated porous medium

$$\boldsymbol{q}_c = \phi(\boldsymbol{v}C - \boldsymbol{D}\nabla C),$$

- $\boldsymbol{q}_c$ is the solute flux vector.
- $\boldsymbol{v}$ is the velocity field.
- $\boldsymbol{D}$ is a tensor that depends on the velocity field and molecular diffusion in the porous medium.

$$\boldsymbol{D} = \alpha_T|\boldsymbol{v}|\mathbf{I} + (\alpha_L - \alpha_T)\frac{\boldsymbol{v}\boldsymbol{v}}{|\boldsymbol{v}|} + wD_0\mathbf{I}$$

Dispersivity vs. water saturation

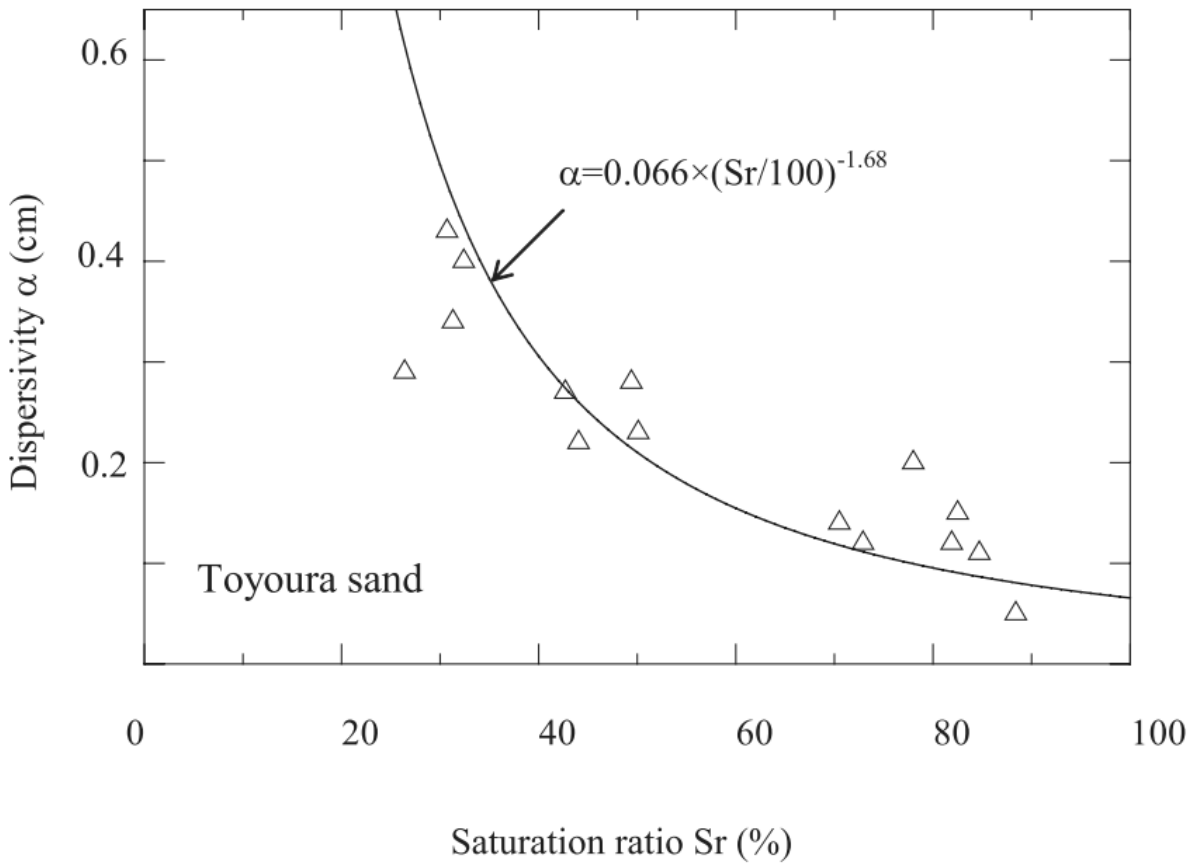


Figure 10. Dispervisity versus saturation ratio in Toyoura sand.

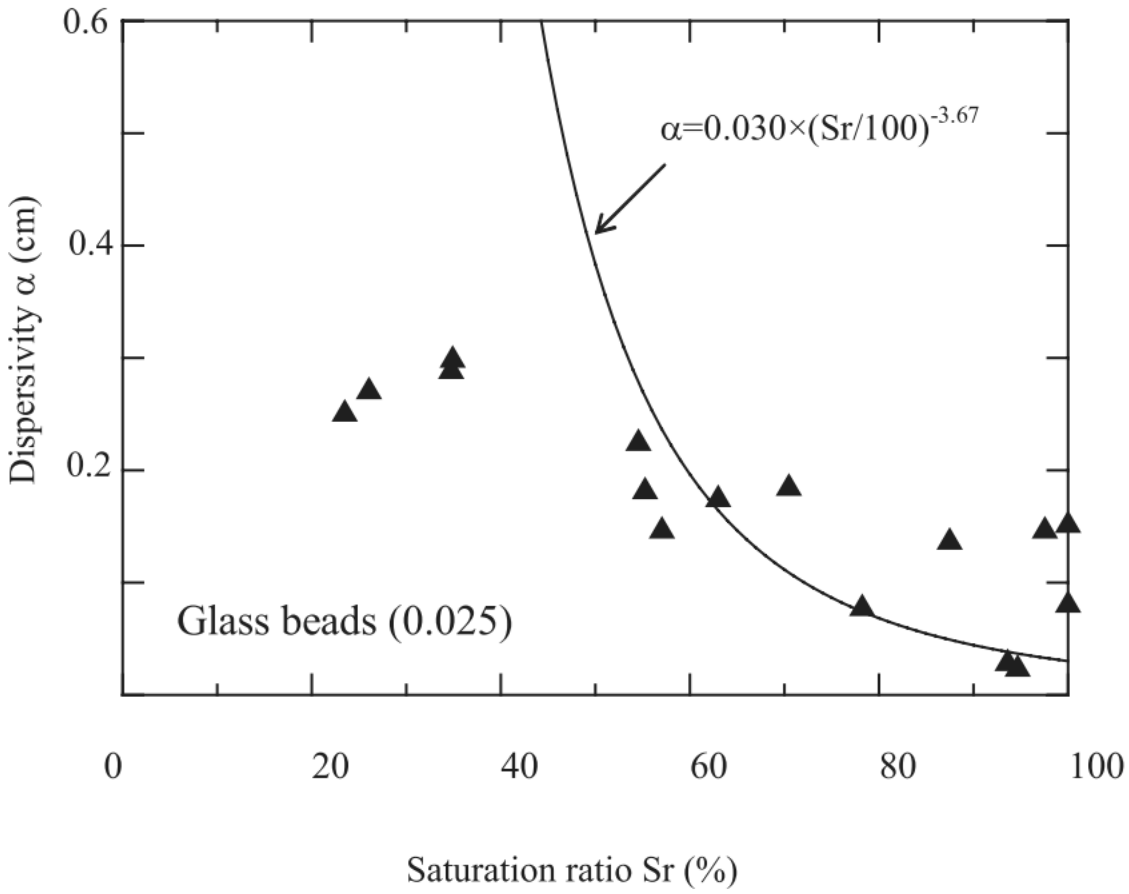


Figure 11. Dispersivity versus saturation ratio in glass beads (0.025).

Sato et al., 2003, WRR

Dispersivity vs. water saturation

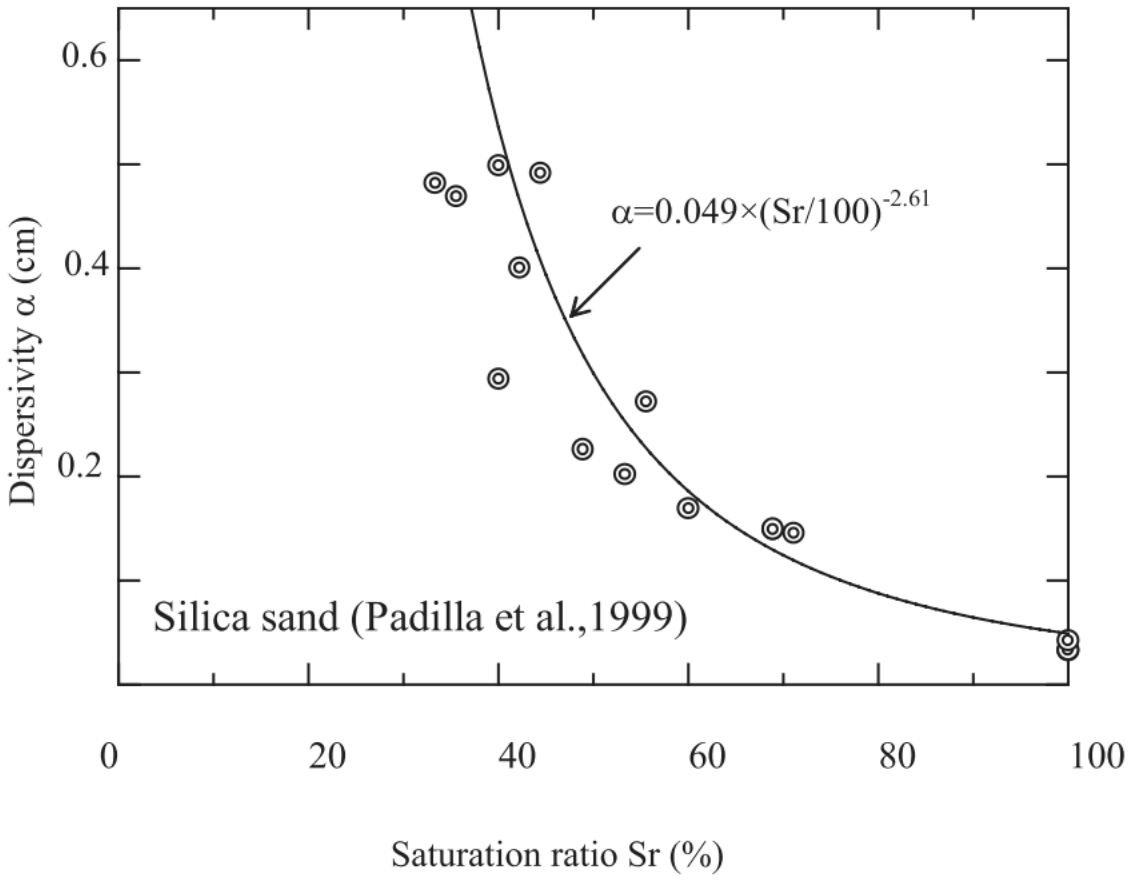


Figure 12. Dispersivity versus saturation ratio in silica sand [*Padilla et al.*, 1999].

Vapor phase transport

In the vadose zone, vapor can migrate upward, laterally, or toward the atmosphere even while water moves downward. The dominant mechanism is site-specific and can shift with moisture content, pore size, and forcing conditions.

Gas Diffusion

Occurs when concentration gradients exist. Usually the baseline mechanism in unsaturated soils. It is typically 4 to 5 orders of magnitude greater than aqueous diffusion.

$$\boldsymbol{D} = w(\theta_a) D_0 \mathbf{I} \qquad w(\theta_a) = \theta_a^{7/3} / \theta_s^2$$

Gas Advection (and dispersion)

Bulk gas flow driven by pressure gradients. Important when permeability is high or extraction / barometric forcing exists. If gas advection is important, then solving the Richards equation alone will not be sufficient, and we need to solve the two-phase flow equations.

Gas components can partition between water and air phases. As a result, their transport can occur through both the aqueous phase and gas phase.